CHEVRON PACIFIC INDONESIA - SUMATERA OPERATION
COMPLETION & WORKOVER PROG. LVL 2
Duri - DURI05720 - 6N-83C

RIG:	BN-11
AFE NUMBER:	2500CP7456
AFE AMOUNT:	$ 68,826
PROGRAM NUMBER:	DURI0572020190104RLN01
EST. TIME:	6 days

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TABLE OF CONTENT

A. OBJECTIVES

B. WELL PROGRAM OBJECTIVE QUALITY EVIDENCE

1. BASIC WELL DATA
2. AS BUILD DATA
    - 2.a. CASING LINER DATA
    - 2.b. FLUID INSIDE CASING
3. COMPLETION INTERVAL
    - 3.a. GEOLOGICAL DATA
    - 3.b. WELL PRESURE, WELL TEMP. CALCULATED MASP
4. DRILLING HISTORY
5. WORKOVER HISTORY
    - 5.a. WELLHEAD & TREE
    - 5.b. WORKOVER JOB SUMMARY
    - 5.c. MECHANICAL & OPERATION ISSUES
    - 5.d. CBL INFORMATION
6. WELL CONTROL RISK
7. COMPLETION FLUIDS MANAGEMENT PLAN
8. BARRIER
9. BOP CONFIGURATION
10. CHOKE MANIFOLD CONFIGURATION
11. SHEAR TABLE
12. BOP TEST PRESSURE & FREQUENCY

C. JOB PROCEDURE

A. OBJECTIVES & BACKGROUND

Program objectives:

1. Maintain Well Control at All Times
2. Provide safe and reliable operations
3. Kill the well.
4. POH Tubing and TP
5. Pull out liner.
6. DOC and COS to PBTD
7. R/T Scrapper
8. Run and Install new liner
9. COS to PBTD
10. TIH TP
11. POP

Program Justification

Finding:
6N83C is CH well and produce oil only from open perforation on R1 sand and have been produce from P1, P2, and K2 sand. Current PBTD is 373 ft MD. This well was drilled until 800 ft MD. in 2007, team decided to raise up PBTD and produce from rindu sand only at that time. Based on TOW, there is remaining oil on bottom P5 and kedua sand. WHT is already Hot, decreasing fluid rate after install meshrite tail jacket in tubing pump.

Recommendation:
POH the pump, clean out sand to 750 ft (new PBTD). Install SASL 0.008 at perforated interval and install regular TP 2.75 inch and set 30 ft above PBTD

Justification:
Based on latest TOW and DTSM model, remaining oil on PK sand with total net pay 110 ft, SO 0.2, HPT 7.7 ft, permeability 1878 md, reservoir temperature 266 degF, and steam thickness 116 ft. Asphaltene: 14 .

B. WELL PROGRAM OBJECTIVE QUALITY EVIDENCE

1. Basic Well Data (WellView)	
Well Grid	6N-83C
Well Name	DURI05720
Well Coordinate	1° 19' 12.098'' N
	101° 14' 33.157 '' E
Well Type	Oil Producer
	Vertical
Spud Date	10-Jun-01
RTE (feet)	61
GLE (feet)	44
MD/TVD (feet)	800 /
PBTD	379

Wellhead, Tree and Surface :

Note:

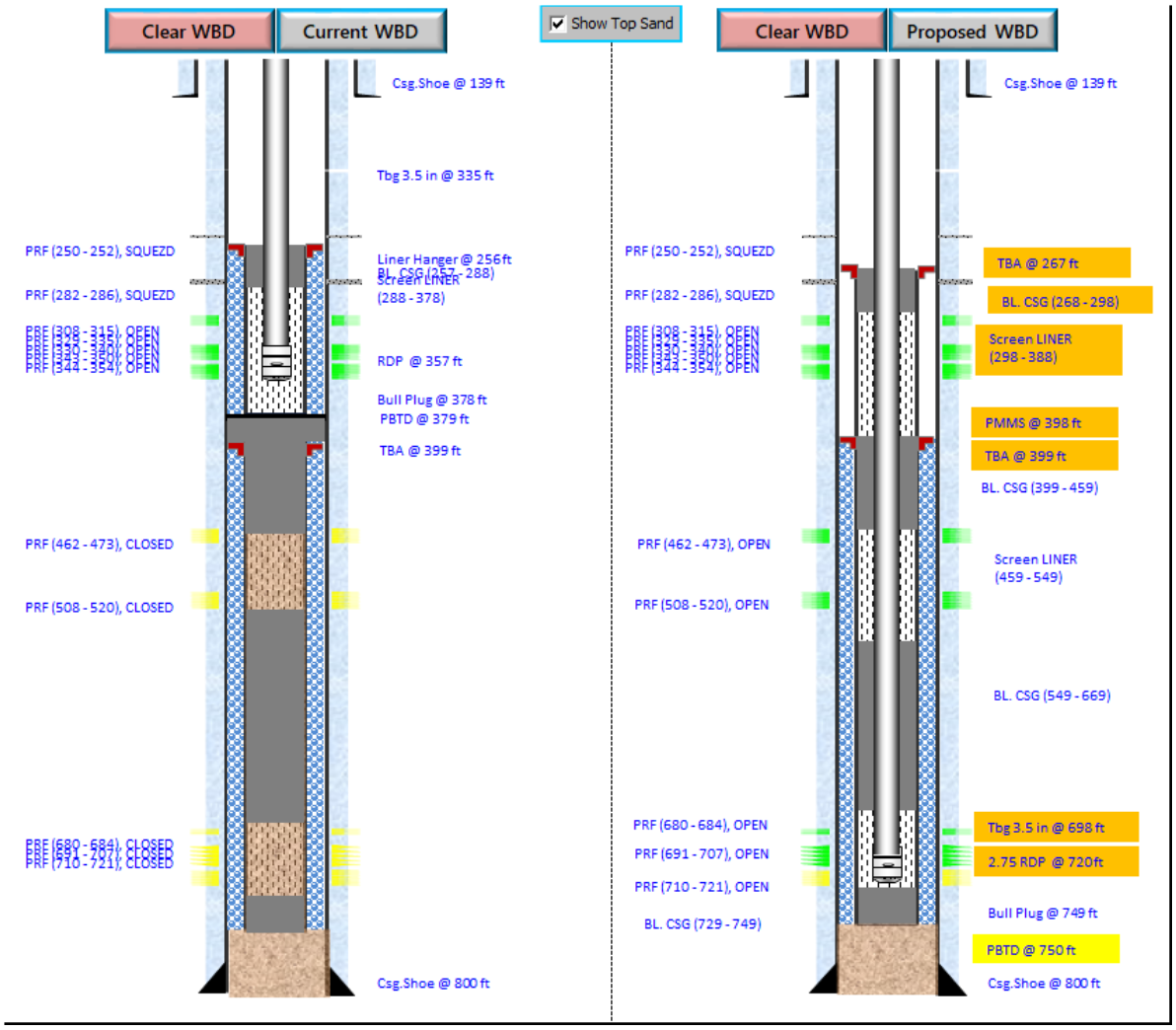
Tubing Head : 10-3/4” SOW x 11” 2M

Tubing Hanger: 11” Nom x 3-1/2” NUE

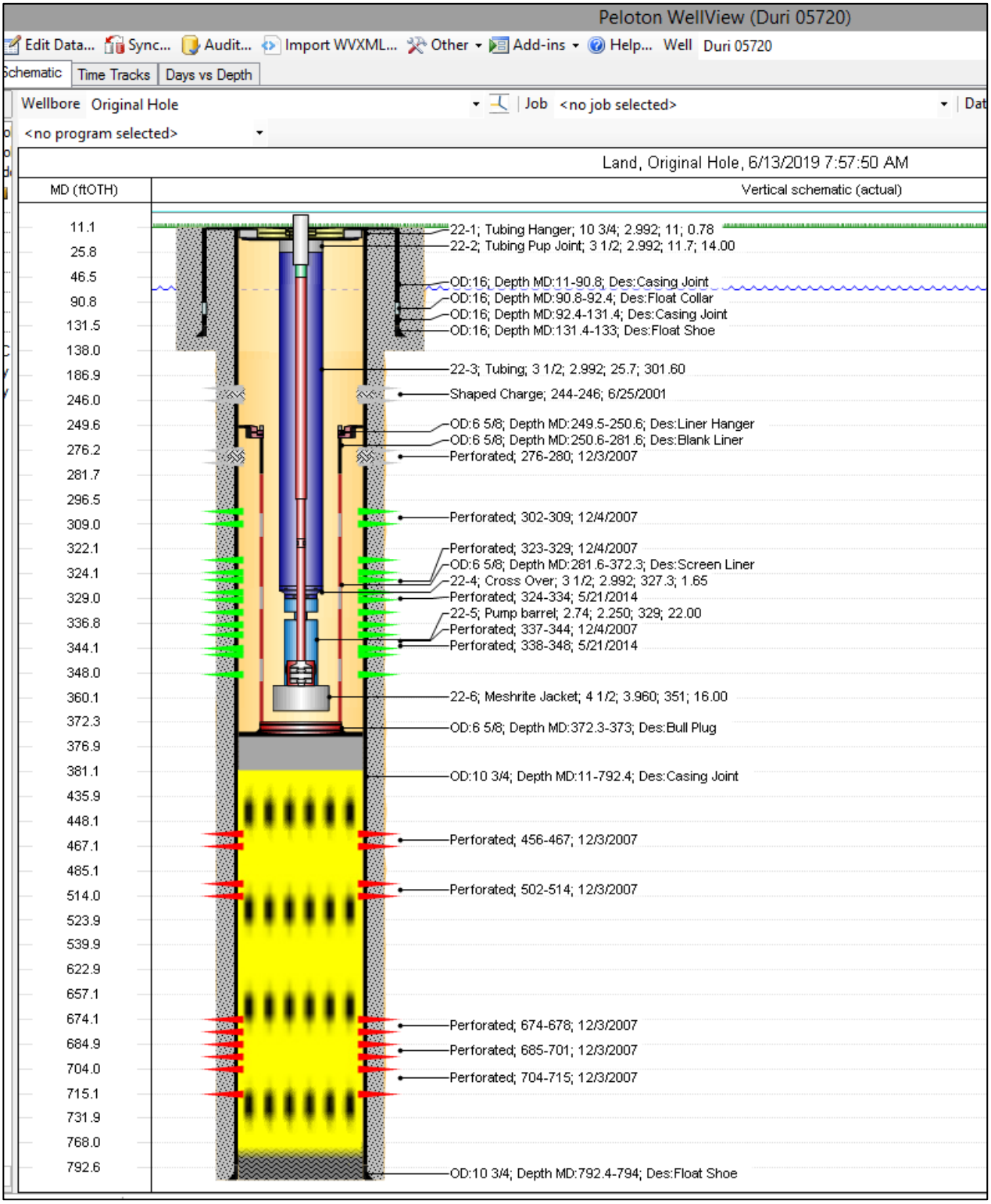
Cover Plate: 11” 2M x 3-1/2” Nom

No issue with wellhead

Before and After Diagram



Well Schematic



2a. As Build Data - Casing/ Liner Data			
Type	Surface Casing		
OD (Inch)	16	70% Burst Rating (psi)	1148
ID (Inch)	15.25	70% Collapse Rating (psi)	441
Weight (lb/ft)	65.0	Tension Rating (Klbs)	736
Grade	K-55	Setting Depth (ft MD)	139
Pressure Test	Not tested	Setting Depth (ft TVD)	139
Type	Intermediate Casing		
OD (Inch)	N/A	70% Burst Rating (psi)	N/A
ID (Inch)	N/A	70% Collapse Rating (psi)	N/A
Weight (lb/ft)	N/A	Tension Rating (Klbs)	N/A
Grade	N/A	Setting Depth (ft MD)	N/A
Pressure Test	N/A	Setting Depth (ft TVD)	N/A
Type	Production Casing		
OD (Inch)	10 3/4	70% Burst Rating (psi)	2191
ID (Inch)	10.05	70% Collapse Rating (psi)	1106
Weight (lb/ft)	40.5	Tension Rating (Klbs)	629
Grade	K-55	Setting Depth (ft MD)	800
Pressure Test	No tested	Setting Depth (ft TVD)	800
Type	Production Liner		
OD (Inch)	6-5/8”	70% Burst Rating (psi)	3577
ID (Inch)	5.92	70% Collapse Rating (psi)	3192
Weight (lb/ft)	24	Tension Rating (Klbs)	382
Grade	K-55	Setting Depth (ft MD)	373
Pressure Test	No tested	Setting Depth (ft TVD)	373
Type	Production Tubing		
OD (Inch)	3 1/2	70% Burst Rating (psi)	4060
ID (Inch)	2.99	70% Collapse Rating (psi)	4004
Weight (lb/ft)	9.2	Tension Rating (Klbs)	109.37
Grade	J-55	Setting Depth (ft MD)	351
Pressure Test	Tested 300 psi Aug 2017	Setting Depth (ft TVD)	351

2b. As Build Data - Fluid inside Casing	
Fluid Type	Oil & Fresh Water
Fluid Weight	7.67 - 8.33 PPG

3a. Completion Interval - Geological Data (Refer to VBWO & UMP - HO page 10)

6N83C

Penetrated	Top Sand	Pore Press. (PPG)		Frac Press. (PPG)		Temp (deg. F)		Hydrocarbon
Sand	TVD (ft)	Min	Max	Min	Max	Min	Max	Content
Rindu	253	4.12	7.81	11.05	11.49	217	257	OIL & STEAM
Pertama	452	2.42	2.94	10.62	11.95	258	283	OIL & STEAM
Kedua	648	2.96	3.12	11.05	12.22	187	278	OIL & STEAM

3b. Completion Interval - Well Pressure, Well Temperature and Calculated MASP

Pressure and temperature refer to	Pressure refers to Rindu 'sd Temperature refers to Pertama 'sd
Max. Zone Pore Pressure (PPG)	7.81 ppg
Min Zone Frac Gradient (PPG)	10.62 ppg
Max. Zone Temperature (deg F)	283 degF

MASP Calculation:

Heavy Oil Avg. API	22	deg.API	
Heavy Oil SG	0.92	SG	Don't change these number.
Heavy Oil Fluid Density	7.677	ppg	
Heavy Oil Fluid Gradient	0.3992	psi/ft	
Gas Gradient	0.1000	psi/ft	
Gas/Oil Mixture Gradient	0.2300	psi/ft	

Reservoir Name	Sand Status	TVD (ft)	Reservoir Pressure (psig)	Res. Temp (deg.F)	HC Content	Fluid Gradient (psi/ft)	MASP (psi)
Rindu	Exposed	253	103.0	257	Oil/Gas	0.2300	45
Pertama	Exposed	452	70.0	283	Oil/Gas	0.2300	-34
Kedua	Exposed	648	106	278	Oil/Gas	0.2300	-44

Note:

MASP during bullheading to be limited to 300 psi

Maximum allowable steam injection in DSF (surround well) is 0.6 psi/ft

4. Drilling History	
Bit & Primary Cementing History	
11-Jun-2001	20 inch bit to 144 ft
10-Jun-2001	20 inch bit to 60 ft
13-Jun-2001	13.5 inch bit to 800 ft
12-Jun-2001	8.5 inch bit to 800 ft

5. Workover History	
Wellhead & Tree :	Tubing Head : 10-3/4" SOW x 11" 2M Tubing Hanger: 11" Nom x 3-1/2" NUE Cover Plate: 11" 2M x 3-1/2" Nom
Workover Job Summary	
End Date	**Review Description**
11-Apr-2019	BDJ 02 - PMP27 - Rod Pump-Bump Down Job – No Well Control Issue SET BDJ TRUCK PT. PRAKARSA - PJM. RUN PRESS 30/20 PSI. BUP W/10 STRKS. PRESS 140/40-80/80 PSI. LOWERED PLGR 6 INCH. FISH, HIT & RESET STD VLV. INJ WTR THRU ANNULUS 30 MNT BY KILLING LINE. BUP W/10 STRKS. PRESS 300/150-300/300 PSI. FINAL PRESS 50/40 PSI. THE WELL RE POP @ 16:00 HRS.
07-Jul-2017	PMP09 - Rod Pump Stuck Normal – No Well Control Issue ▪ CHECKED SITP: 20 PSI, SICP: 15 PSI, COOLING DOWN W/ PUMP RATE: 1 BPM, TTL 60 BCW, BLEED OF PRESSURE TO 0 PSI, OBSERVED SITP & SICP =0, CONFIRMED WELL DEAD. ▪ TOH W/1-3/4" PLUNGER ON 8 JTS 3/4" S.ROD (USED 3/4" SHEAR COUPLING) . ▪ RELEASED W/ 10-3/4" D.NUT + FOSV AND LOWERED 1-3/4" TBG PUMP TO CHECK TOS, TAGGED @ 360' (19' SAND FILL). ▪ FOAMING OUT SAND F/360' TO 379', OBSERVED SAND ON RETURN. ▪ TIH 2-1/4" TBG PUMP (10-52626) ON 10 JTS 3-1/2" NUE TBG + 14' 3-1/2" NUE PUP + 1 JT 38' 4-1/2" MESHRITE JACKET, PUMP SET 357.74'. ▪ TIH W/ 2-1/4" PLUNGER ON 2 JTS 3/4" S.ROD (USED 3/4" SHEAR COUPLING) + 10 JTS 7/8" S.ROD + 14' 7/8" PONY ROD + 1-1/4" POLISH ROD. PLGR SET @ 357'. ▪ PRESS TEST PRODUCTION TREE (STUFFING BOX, SAMPLE COCK, MASTER VALVE & ALL TREE CONNECTION) TO 650 PSI (500 PSI + MASP) F/ 5 MNTS ALONG W/ TEST TP RECORDED THE TEST ON BARTON CHART. TESTED PUMPING W/300 PS
06-Dec-2007	REC11 - Recompletion Rindu Producer – No Well Control Issue • TAKE OUT 10-3/4 INCH TBG DOUGHNUT AND LOWERED 2-1/4 INCH TBG PUMP TAG AT 725 FEET. • DUMP SAND W/ 20/40 MGPS TOTAL 132 SXS. • M/U AND RIH W/ 10-3/4 INCH CSG SCRAPPER ON 12 JTS 3.5 INCH TBG, TAG AT 387 FEET

	• CEMENT CAP F/ 387 TO 379 W/ 5 SXS CMT + 2 SXS SILICA FLOOR + DISPLACE 2.5 BBLS. W.O.C CEMENT CAP • R/U WIRE LINE AND HLS RUN CBL TO CORRELATION TAG AT 389 FT FOUND BAD BONDING DECIDED TO PERFORATE IB INTV 282 - 286 • ASSEMBLED W/ ON 3 JTS 6-5/8 INCH X 30 FT S. LINER + 6-5/8 INCH B. PLUG + 1 JTS 6-5/8 INCH X 30 FT BLANK CSG , RIH W/ 4 JTS 3.5 INCH EUE TBG STINGGER + 10-3/4 INCH GP TOOL. RIH 6-5/8 INCH S. LINER • RIH W/ 6-5/8 INCH S. LINER + 10-3/4 INCH GP TOOL ON 8 JTS 3.5 INCH DP. SET BTTM B. PLUG AT 379 FT, TBA AT 256 FT, GP TOOL AT 251 FT. • RIH W/ 6-5/8 INCH SETTING NUT + 8 JTS 3.5 INCH DP, SET TO TBA AT 254 FT. • RIH W/ 2-1/4 INCH PLGR ON 1JT 7/8 INCH S.ROD + 7/8 INCH SHEAR COUPLING + 11 JTS 7/8 INCH S.ROD + 3 PCS 7/8 INCH X 6 FT P.ROD + 1 PC 7/8 INCH X 2 FT P.ROD + 1/1/4 INCH PLSH ROD, PLGR SET. 358 FT. • RIH W/ 2-1/4 INCH T.P(10-51021) ON 10 JTS 3-1/2 INCH NUE TBG + TBG HANGER, P.SET. 358 FT, FILL UP STRING AND PRESSURE TEST W/ 200 PSI
29-Jun-2001	COM02 - Initial Completion Oil Producer – No Well Control Issue • RIH W/10-3/4" CSG SCRAPPER ON 26 JTS 3-1/2" DP TO 774' PBTD. PREPARING TO POH 10-3/4" CSG SCRP. • ASSEMBLED W/6-5/8" BP + 10 JTS 6-5/8" X 30' SL + 1 JT 6-5/8" X 20' SL + 1 JT 6-5/8" X 30' BLK CSG + 1 JT 6-5/8" X 20' SL TAIL TALE + TBA + 11 JTS 3-1/2" NUE TBG STINGER + PUP TBG + GP TOOL. • RIH W/6-5/8" SL (0.012") ASSY ON 13 JTS 3-1/2" DP. TOP TBA @ 339', TOSC @ 451', BTTM STINGER @ 766', BP @ 774'. • RIH W/10-3/4" L.SEAL ON SWG TOOL + 4 JTS 4-3/4" DC + 9 JTS 3-1/2" DP SET @ 399'. • RIH W/OE ON 11 JTS 3-1/2" TBG + 14 JTS 3-1/2" DP, TOS @ 769' • TESTED & RIH W/2-3/4" TP (10-61563) + 22 JTS 3-1/2" NUE TBG + 10-3/4" D.NUT. PUMP SET @ 749'.

Mechanical & Operation Issues :

Date	Description
N/A	N/A

6. Well Control Risks *(Refer to VBWO & UMP – HO Document page 12)*	
Reservoir Pressure	• Risk having well kick due to insufficient Hydrostatic Pressure. • All Injectors connected to the well had been reviewed • Pressure may be higher than anticipated due to Steam • All injectors impacted to this well has been reviewed. • The offset injector that need to be shut in prior to rig execution and to be monitored is mentioned in Offset Injector Wells Management Plan.

<table>
<tr><td></td><td>No killing problem history
MASP from exposed zone Rindu ‘sd is 45 psi.
Maximum pressure from Rindu ‘sd is 7.81 ppg
Please anticipate any KCL and polymer to encounter lossess or high density fluid as necssary</td></tr>
<tr><td>Reservoir Fluid</td><td>Steam breakthrough causing personnel injury to fatality
Higher GOR causing high shut-in pressure and flammability
Presence of H2S causing personnel injury
Ensure Gas Detector is in place and functioning
Hot water or crude oil causing personnel injury</td></tr>
<tr><td>Fluid Losses</td><td>Kick due to loss of hydrostatic barrier
Inadequate LCM
In every clean out sand procedure, rig use support from foam unit (foaming job or aerated polymer).</td></tr>
<tr><td>Mechanical Condition of Wellbore</td><td>Plunger and rods left in hole unable to recover
Tubing stuck, unable to pull with maximum capacity
Tubing broken/ collapsed after pulling
Sand/ clay fill up wellbore, blockage the annulus</td></tr>
<tr><td>Trapped Pressure under existing barrier</td><td>Trap pressure below tubing hanger. However, likelihood is very low and always ensure well stable (0 psi on surface) prior to pull Tubing Hanger
Pressure trap below Standing Valve
Pressure trap below sand/ clay that blockage annulus</td></tr>
<tr><td>Presence of HC formations nearby and hydraulic isolation</td><td>Loosing hydraulic isolation can cause Well Control Situation, causing personnel injury to fatality, environment damage, cost and loss of company reputation</td></tr>
<tr><td>Offset Injector Wells Management Plan</td><td>Request to shut in following injector well prior to rig to commence.
N/A
List of surrounding injector wells that may potentially impact to the wellbore are:
<table>
<tr><th>NO</th><th>WELLNAME</th><th>SAND</th><th>DISTANCE (m)</th></tr>
<tr><td>1</td><td>6N82B-R</td><td>R</td><td>96</td></tr>
<tr><td>2</td><td>6N62B-R</td><td>R</td><td>206</td></tr>
<tr><td>3</td><td>6N71A-L</td><td>P</td><td>304</td></tr>
<tr><td>4</td><td>6P11B-R</td><td>R</td><td>329</td></tr>
</table></td></tr>
</table>

	No need to shut-in those injectors due to no well control issue in history. But D&C will request to Asset Team if there is killing problem encounter during execution.

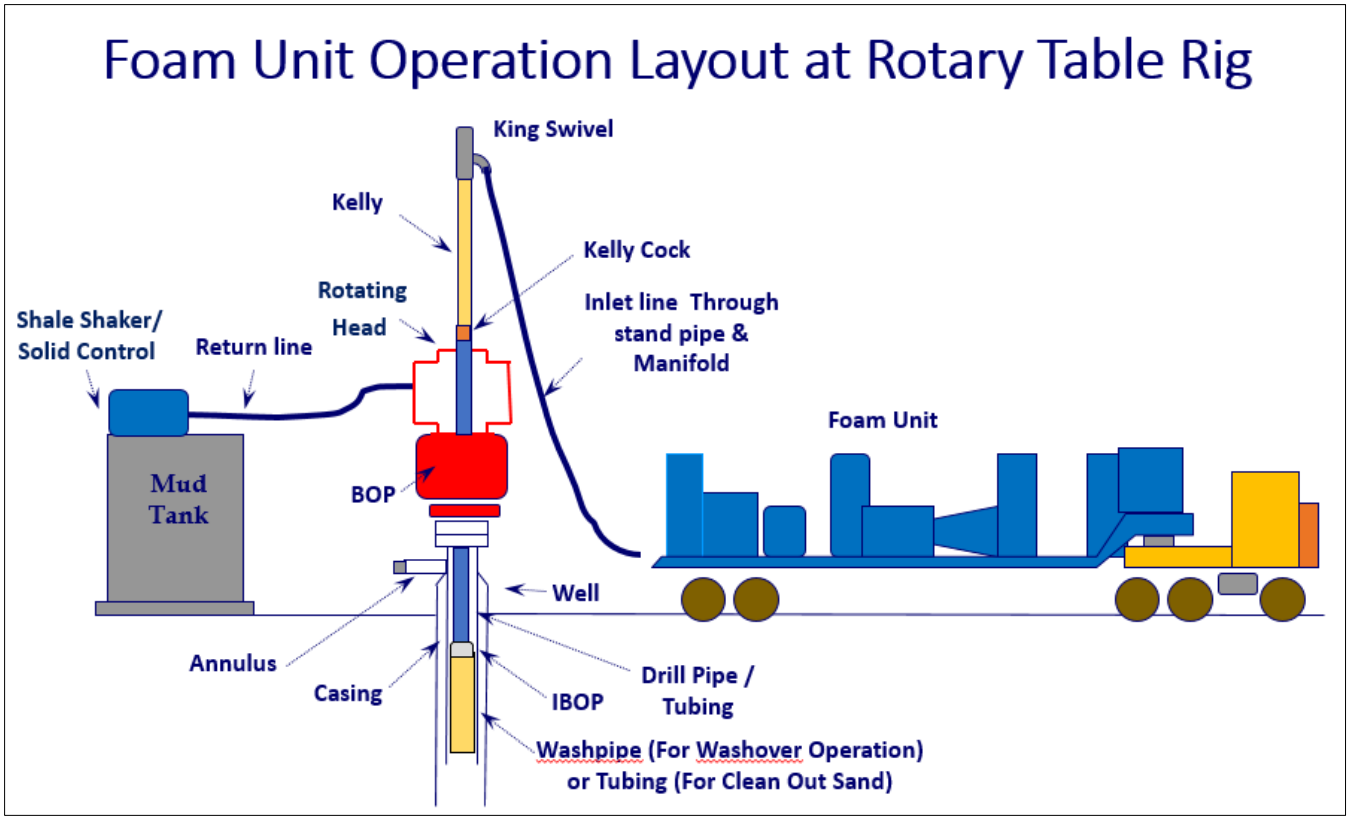
7. Completion Fluid Management Plan *(Refer to Well Design & Plan Guideline for HO, page 10 - 12)*	
Min Overbalance	- Initial Killing use Fresh or Formation water (8.33ppg). Fill up well to cool down and maintain over balance. - If water density is not enough to kill the well then completion fluid density can be determined as following formula (ensure well's annulus or string has been filled up with fresh or formation water) : KF ppg = 8.33 ppg + {(Shut-in Pressure*)/(0.052 x TVD)} + 0.5 ppg (SF) * If bullheading through tubing use SITP If bullheading through annulus use SICP - Completion Fluid Brine i.e. KCL, NaCl and combination of Sodium Formate and Sodium Nitrate are commonly used for killing fluid within the area operation which will be provided by a third party Contractor - We will use safety factor 0.5 ppg for any kill fluid.
Fluid availability	- Each of area has dedicated stand pipe or central gathering station (either treated formation water or fresh water): Stand pipe 5U and 6L Brine and Mud are provided by third party contractor (PT. Matra Unikatama). - This well has cased hole completion with 10-3/4" production casing. One bottom up volume is 80 bbls. (Assuming wellbore ID 10.05" from surface to 794 ft) - There are 2 tanks available in rigsite with both have maximum 200 bbls capacity or total for 2 tanks are 400 bbls. With average pump rate 2-5 bpm, 2 tanks could supply water for 80 – 200 minutes depends on the fluid loss and supply water from killing line. - If killing line doesn't have enough fluid rates, rig will use vacuum or water tanker and take supply water from 5U/6L stand pipe. - With 1 load of vacuum or tanker capacity is 60 bbls, we need 6.6 load to full the tank. - We need the vacuum to re-fill up rig tank every 12 – 30 minutes to get continues pumping (assuming no fluid in the tank). - Based on assessment, the access road to location is good enough for vacuum truck to reach the location. - In pre-spud meeting, we will discuss the availability of vacuum truck with operation team.
Loss Circulation	- General guideline: Please follow Loss Circulation SOP page 11-12 https://collab001-idr.sp.chevron.net/sites/smodnc/smodncws/WellSafe%20Document/Loss%20Circulation%20SOP%20Rev%202.pdf

	- 1st Plan to use XCD polymer starting from 40 SMF until maximum 80 SMF if found any loss during clean out sand, if not succeed please move to 2nd plan. - 2nd Plan is to use Aerated XCD Polymer until got fluid return. if not succeed please move to 3rd plan - 3rd Plan to use foam if found any loss during clean out sand
TCT/PCT	N/A
Hydrates	N/A
Spike Fluid	- Killing Fluid for normal operation shall be on location. However, in case need a higher density fluid, it shall be ready on Service Company yard where located within 30 Km from DSF location

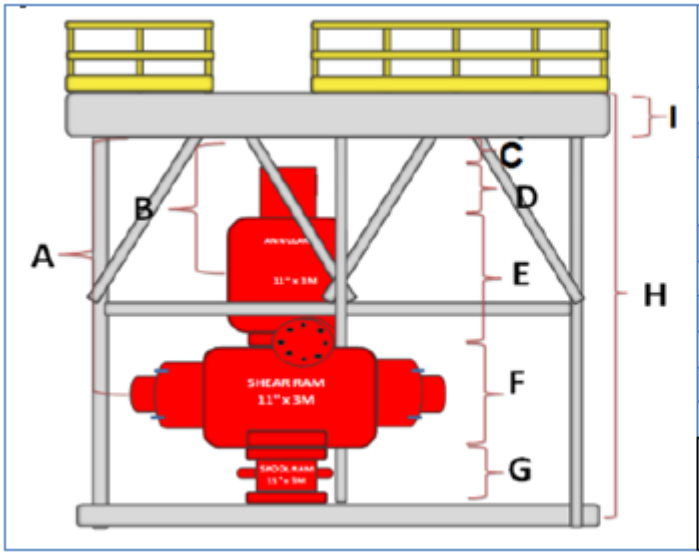
8. Barrier Summary *(Refer to Well Design & Plan Guideline for HO, page 21 - 22. The detailed is on Section C Procedure)*		
Operation	**Internal**	**External**
N/U & N/D Tree	Hydrostatic	Hydrostatic & Tubing Hanger with seal mechanism
N/U & N/D BOP	Hydrostatic & FOSV	Hydrostatic & Tubing Hanger with seal mechanism
Running String (Tubing/ DP)	Hydrostatic & FOSV/IBOP	Hydrostatic & BOPE
Displacement/Circulating/Foaming	Hydrostatic & FOSV or IBOP	Hydrostatic & BOPE
Washover	Hydrostatic & FOSV/ IBOP	Hydrostatic, Kick Control Joint & BOPE
Running Screen Liner	Hydrostatic & FOSV/ IBOP	Hydrostatic & BOPE with Kick control joint is on rig floor
Running Rod	Hydrostatic & FOSV	Hydrostatic & Tubing Hanger with seal mechanism

8 B. Displacement Procedure *(Refer to Well Design & Plan Guideline for HO, page 17 - 21. The detailed is on Section C Procedure)*

If there is a foaming job on this well then internal barrier are hydrostatic, Kelly Cock/FOSV and IBOP while external barrier are hydrostatic and BOP. BOP will be in open position during the displacement. Kelly Cock/FOSV and IBOP will be installed at work string during execution to secure work string in case well control incident happen. Detail procedure for Aerated/foaming job is available at section C Procedure.

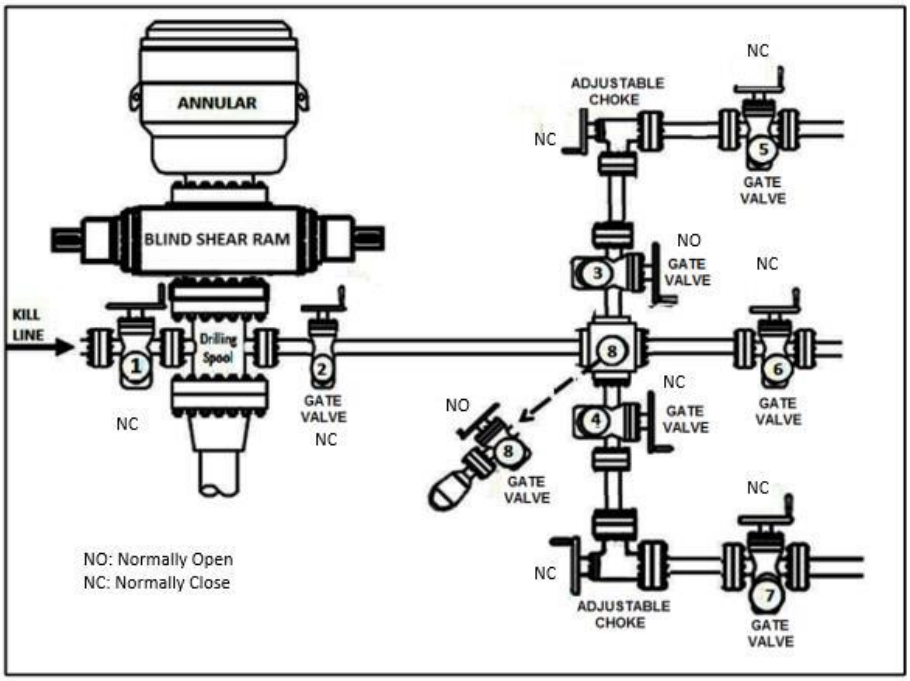


9. BOP Configuration		
RIG : BN-11		
No	**Acceptance Criteria**	**Data**
1	Quantity, Bore ID & Pressure Rating of annular Preventer (s)	1 (ea), 11" 3000 psi
2	Quantity, Bore ID & Pressure Rating of the Ram Preventer(s)	1 (ea), 11" 3000 psi
3	Configuration of Ram Preventers to include position within the stack and Ram Type (I.e. Fixed Pipe, Variable Bore, Casing, Blind Rams, Blind Shear Rams, Shear Rams, Casing Shear Rams, etc.)	See picture BOP configuration below
4	For each Pipe Ram type preventer provide pipe size or size range and hang off capacity	Single ram, size 11” x 3000 Psi c/w Blind Shear Ram, See picture BOP configuration below
5	Location, bore ID and Pressure Rating of the choke and kill valves & lines	3-1/8 Inch x 3000 Psi (Choke), 3 Inch x 3000 Psi (pipe line) See picture Choke Manifold configuration below
6	Reference point and distance from reference for each component in the stack	See picture BOP configuration below
7	Temperature rating and fluid compatibility of BOP rubber/ elastomers	Natural or Nitrile Rubber with temperature 20 degF – 190 degF, compatible with normal fluids in HO Operation Average surface temperature at HO is between 95-110 degF https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/General%20OQE/Clarification%20Temperature%20Surface%20at%20DSF.msg



	Component Description	Spacing ft
A	CTR RAM to BTM RT	10.25
B	CTR ANNULAR to BTM RT	7.83
C	SPACE (TOP RH-BTM BEAM RT)	3.25
D	ROTATING HEAD	3.08
E	ANNULAR	3.25
F	RAM	1.92
G	DRILLING SPOOL	1.5
H	BTM DS – TOP RT	16.00
I	BEAM RT	3.00

10. Choke Manifold Configuration



SERIAL NUMBER	G1505655
VALVE SIZE	3-1/8 INCH
RWP	3000 PSI

11. Shear Table

SHEAR CAPACITY							
BL FZ 11", 3000 psi WP, Single Shear Ram Bope - Heibei Bo Lu Tianbao							
Tubular :	2-3/8", 4.70 lb/ft	2-7/8", 6.50 lb/ft	3-1/2", 9.3 lb/ft	3-1/2", 9.5 lb/ft	4-1/2", 12.6 lb/ft	4-1/2", 15.1 lb/ft	5", 19.5 lb/ft
Grade	L-80	J-55	E-75	G-105	K-55	G105	G-105
No Well Bore Pressure	1541 psi	1685 psi	1712 psi	1998 psi	1986 psi	2012 psi	2031 psi

12. BOP Test Pressures and Frequency			
Equipment	Pressure Rating (psi)	Allowable Pressure Rating (psi)	Tested
Annular* (see note)	3,000	2,100 (70%)	In Procedure
Ram* (see note)	3,000	3,000	In Procedure

Note:

1. Low Pressure test BOP shall be conducted with 250 - 350 psi and for High Pressure Test BOP shall be conducted with 650 psi or minimum 500psi + MASP (Ensure not to exceed 70% Casing Burst Pressure or 100% Wellhead RWP of 70% Annular RWP or 100% Packer RWP

2. BOP Equipment shall be tested as follows:

   a. Not exceed interval of 21 days
   b. Before the equipment is put into operational service on wellhead
   c. In accordance with the equipment owner PM program
   d. Following the disconnection or repair of any wellbore pressure seal in the BOP stack assembly or choke manifold (limited to the affected components only)

C. WELL PROGRAM

GENERAL PROCEDURE FOR WORKOVER HOOU - IBU SMO COMPLETION

WELL NAME / ID	**6N83C-01**
JOB TYPE	**RLN01 - Revise Liner**
CURRENT WELL STATUS	**ON**
PROGRAM NUMBER	**DURI0572020190104RLN01**
WELL PROFILE	**VERT**
PROD CASING SHOE	**800 ft**
CURRENT TBA	**256 ft**
CURRENT/ PROPOSED PS	**357 ft / 720 ft**
CURRENT PBTD	**379 ft**
DEVELOPED BY	

Last Job History:

- 6N-83C is CHGP well and currently produce oil from Rindu sand and was drilled on June 2001 with original PBTD @774'
- 6N83C have been produce from P1, P2, and K2 sand. Current PBTD is 373 ft MD.
- Raised up PBTD on Nov 2007 by dump sand and cement cap from 774' to 373' (REC 11)
- Based on latest TOW and DTSM model, remaining oil on PK sand with total net pay 110 ft, SO 0.2, HPT 7.7 ft, permeability 1878 md, reservoir temperature 266 degF, and steam thickness 116 ft. Asphaltene: 14 %.
- Last Wellservice PMP09 job was on Jul 2017 using CPI 77
- No killing problem was found during driling job or work over job
- Sand fill history 10-30 ft
- Loss potential, Please prepare foaming unit to perform FOS and please be aware during operation (potential cross flow from rindu sand to PK sand)

Head Up List before moving into the well:

- Please order 2-3/4 TP regular
- Please order 0.008" Screen Liner and Centralizer
- wvd Readiness
- Please ensure to isolate power source by FOD

No	PROCEDURE	Time (hrs)
1	**MIRU RIG.**	12
2	**RECORD SICP AND SITP, RECORD IN THE MORNING REPORT.** **LINE UP LINES FROM CASING AND TUBING TO TANK. BLEED OF CASING AND TUBING PRESSURE PRESSURE TO 0 PSI (OR AS LOWEST AS POSSIBLE). CONTINUE TO KILL THE WELL.** **Make sure to kill the well properly before commencing workover operation and also maintain primary well Control at all time (WSEA 3A and 3B)** **Last Successful Killing: Water 8.33 ppg with volume max 60 bbls.** **Note: for well with killing problem history (RLN12) we may follow last successfull killing recipe above or may use other alternative based on actual condition. Please discuss with fluid engineer and team leader operation for more detail information.** Fresh or Formation water (8.33 ppg) is used to fill up wells in most location in HO due to depleted formation. Filling up well with water is normally performed to maintain fluid return to surface and maintain over balance. If the well unable to maintain fluid column, operation can be continued while pumping fluid in to the well regularly If water density is not enough to kill the well then completion fluid density can be determined as following formula (ensure well's annulus or string has been filled up with fresh or formation water) : KF (ppg) = 8.33 (ppg) + 0.5 ppg (SF) + (Shut-in Pressure (psi)*)/(0.052 x TVD top of perforation (ft)) *If bullheading through tubing use SITP If bullheading through annulus use SICP	6

	Completion Fluid Brine i.e. KCL, NaCl and combination of Sodium Formate and Sodium Nitrate are commonly used for killing fluid within the area operation which will be provided by a third party Contractor	
3	**DISCONNECT STUFFING BOX, RELEASE POLISH ROD, POH PLUNGER AND SUCKER ROD STRING. IF PLUNGER STUCK GENERATE FISHING TREE.** ***Last Total Rod String:*** ***2-1/4" PLUNGER ON 2 JTS 3/4" S.ROD (USED 3/4" SHEAR COUPLING) + 10 JTS 7/8" S.ROD + 14' 7/8" PONY ROD + 1-1/4" POLISH ROD. PLGR SET @ 357'.*** ***Left in hole:*** **N/A** Fishing Tree Plan : --> Option #1 : Apply over pull and reciprocate plunger or break shear coupling. --> Option #2 : Blind backoff sucker rod. --> Option #3 : Cut sucker rod using rod cutter. --> Option #4 : Pull with tubing pump to recover the succer rod (stripping job) Other options will be dictated based on the fish condition. Discuss with fisherman and TL Operation.	2
4	**N/D X-MAS TREE, INSTALL FOSV, AND N/U BOP (ENSURE WELL SECURE). PERFORM FUNCTION TEST BOP AND CONT W/ PRESSURE TEST BOP'S TO 250 - 350 PSI (LOW PRESSURE TEST) AND 650 PSI (500 PSI+ MASP for HIGH PRESSURE TEST) FOR 5 MINUTES EACH. RECORD ON BARTON CHART.** **MASP = 45 psi** Inspect wellhead connection, valves, seals. replace if necessary.	6
5	**POH EXISTING TUBING STRING** Last Total Tubing String: 2-1/4" TBG PUMP (10-52626) ON 10 JTS 3-1/2" NUE TBG + 14' 3-1/2" NUE PUP + 1 JT 38' 4-1/2" MESHRITE JACKET, PUMP SET 357.74'.	36

Left in hole: N/A

Fishing Options:

1. Please reciprocate and overpull stuck TP.
For 2-3/8" NUE tubing max overpull to 40 klbs, for 2-7/8" NUE tubing max overpull to 60 klbs and for 3-1/2" and 4-1/2" NUE tubing overpull to 80 klbs.
2. R/U Wireline and RIH Jet Cutter. RIH FPI as necessary to determine stuck point - as deep as possible. If Stuck Point 200 ft above TBA, cut tubing at 2 spots (1st in the middle of stucked fish, and 2nd above stuck point).
3. Continue with wash over using rotary shoe and circulate out (if loss use foam unit). For size of rotary shoe please find in reference.
4. Fish out remaining fish (if thread in good condition, screw in tubing to TOF or use die collar or tapper tap, if thread is not good RIH skirt mill and then use Overshot or Spear). For size of overshot and spear please find in reference
5. Repeat step no 2, 3, and 4 as necessary
Please confirm to CE prior performing step no 2, 3, 4 and 5.

Note,

1. If TOF is not clear please Run IB, next job will be decided based on IB result.
2. Do not conduct back off tubing for WPF Rig (if need to back off tubing, please suspend job and move out rig)
3. Please do inspect pulled tubing string condition (every tubing, nozzles, mandrel and cup packer). Report for any anomaly
4. Notes in the tour report if there is any anomaly such as fluid cut, sand fill, scale, etc
5. Please discuss with Fisherman and TL Operation if any deviation.

In case Rig did not successful to pull out stuck tubing pump and have to pending well, make sure to TIH tubing pump (instead of OE tubing) so Rig still able to perform pressure test Xmas tree (all tree, connections, stuffing box, or sample cock whichever applicable) to 500 psi + MASP as per stated in WSEA 10-L

Please find the fishing tree below for reference

Fishing in Casing
https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/General%20OQE/Fishing%20Tree%20Plan%20-%20Casing%20Size%20-%20Rev0.pdf?d=wb475327931ab45c5a0814a2e02bd5d5f

Fishing in Liner
https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/General%20OQE/Fishing%20Tree%20Plan%20-%20Liner%20Size%20-%20Rev0.pdf?d=wfe67e2cb2d9b4855844777edef984fb0

Please find the fishing BHA and condition for reference

	Fishing Tool per hole size *https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/General%20OQE/fishing%20tool%20per%20hole%20size.docx* *Fishing Process with possible condition and BHA Reference* *https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/General%20OQE/Fishing%20process%20Condition%20with%20BHA%20refrence%20in%20HO.xlsx*	
6	**ROUND TRIP WITH CASING SCRAPPER TO LEAD SEAL** Adjust the number of scrapper need to be run based on the well configuration and production zone. If there is indication casing damage, please execute "Contingency Plan for Casing Leaking/ Clay Problem" - Please call TL Operation and CE before conducting this plan.	4
7	**POH LEAD SEAL AND PREPARE TO WASH OVER SCREEN LINER AS PER AVAILABLE SOP. IF SOP IS NOT AVAILABLE THEN PLEASE GENERATE FISHING TREE.** **Fishing Tree Plan:** 1. Fish Out Lead seal with Setting Nut 2. Do Wash Over with Rotary Shoe + Wash Pipe **Preparation:** 1. Run Wash Pipe & Rotary Shoe with correct size & type-dimenssion of rotary shoe (please discuss with Fisherman) 2. M/U with proper torque recomendation (refer to SOP) **Wash Over BHA General:** Rotary Shoe + Wash Pipe + Safety Joint + DC + DP Recommend to Run Wash Pipe step by step (short wash pipe (3 jts) to maximize wash over process), ***not recommend to Run Wash Pipe in a long string assembly*** Please discuss with fisherman to determine BHA as conditional **Critical Point:** 1. Prefer to use wave type of rotary shoe. Short tooth or ocean wave shoes are used in medium and hard formations Longer toothed shoes are used in soft formations	4

	2. Not exceed from recomendation Torque during wash over (3000 - 4000 lbs on DC) 3. Recomendation for Running wash pipe is 300 ft, please discuss with Fisherman	
8	**WASH OVER UPPER SCREEN LINER PROCESS:** 1. Do circulation with clear fluid with reccommendeded velocity 120 ft/min. 2. Do Wash Over with recomended torque & WOB 2. Wash Over with KCL 2% & Polymer 45 SMF 3. If got loss , do : - Raise up to shoe - Increase viscosity of Polymer - Spot polymer, wiper trip & continue wash over with low RPM 4. Use aerated polymer if necessary (follow SOP) 5. After wash over to TD, do reciprocate & clean out till clean - For wash over with aerated, bullhead water via annulus to clean out aerated effect and spot HiVis Polymer via string before pull out string. - Do Spot Polymer and reciprocated string to avoid string stuck - Do Spot Polymer minimum Open Hole Section to kill aerated effect	36
9	**FISHING SCREEN LINER :** 1. Use Setting Nut if no dragging, overtorque, well static during wash over 2. Use Spear with Jar if got dragging, overtorque and well loss during wash over **Note for Open Hole Well:** If Screen Liner to Long > 300 ft, recommend to do into 2 steps (above shoe, below shoe or 2 segment) 1. Wash Over & Cut with Hydraulic Cutter for upper section, continue fish out upper section 2. Wash Over & fish out Lower Section Please discuss with fisherman, WSM Specialist, Team Leader & Completion Engineer	36
10	**RUN ROCK BIT to from 379 ft to New TD at 774 ft**	6

	- 3-1/2” DP (adjusted) - 4-3/4” DC (4-6 joints) - 5-3/4” or 6-1/8” rock bit BHA assembly can be adjusted. Please discuss with team leader operation. 1. Use KCL (2%) + Polymer (min. 45 SMF) 2. Ensure water supply is continue Follow SOP to run rock bit. Please make sure to have fluid return on surface by using appropriate fluid Report if any deviation during running rock bit	
11	**ROUND TRIP WITH CASING & LINER SCRAPPER TO NEW PBTD AT 774 ft** **10-3/4” Casing scrapper to estimate PK TBA at 399 ft** **6-5/8” Liner scrapper from 399 to new PBTD at 774 ft** Adjust the number of scrapper need to be run based on the well configuration and production zone. If there is indication casing damage, please execute "Contingency Plan for Casing Leaking/ Clay Problem" - Please call TL Operation and CE before conducting this plan.	2
12	**RUN PRESSURE WASHER TO JET WASH PK LINER**	2
13	**CLEAN OUT SAND TO 774 FT** RIH with mule shoe and clean out sand to PBTD. Use Foam whenever required if there is experience loss circulation. Ensure water supply is enough during clean out Note, Monitoring displacement process is needed to develop according to potential well control event while displaced water / brine with lower density fluid such foam. Below is procedure for foaming which contain of monitoring parameter and Kick mitigation. 1. Spot Foaming Unit	6

	2. Rig Up Foaming Unit 3. Fill up water tank 4. Set Up line , conduct pressure test on discharge line of foaming unit on 1000 Psi 5. Add and mix the chemical (Foamer) into the water tank depend on wellbore requirement 6. Run in hole and lower the mule shoe / open ended/ rotary shoe and tubular Bottom hole assembly until 2-10 ft above top of sand 7. Install the inside BOPE 8. Conduct the foam / aerated pumping a. Pump the water using triplex pump on 1-3 BPM while monitor pressure gauge and control panel b. Pump the air slowly on 1300 – 1800 RPM while monitor pressure gauge and control panel c. Closed monitor the return (adjust the chemical foam until stable on return) d. Lower the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly slowly while monitor the return • If the tubular string does not utilize the / check valve/ float sub, conduct fill up the wellbore to achieve slightly overbalance condition before disconnect the current tubular for make up the next additional tubular string. • In the event that the return is stop (No Return on surface) , inform the driller to rise up the tubular into the safe area above the open interval / perforation • If foaming job is suspended for certain time, ensure to immediately secure the tubular string into the safe area above the open interval/perforation e. Continue the foam pumping and clean out until PBTD (clean well) , take sand sample from the wellbore on line manifold rig 9. Avoid environmental pollution by Spray the foam on breaker tank using de-foamer to reduce the enormous volume of foam on the breaker tank 10. After well confirm on clean condition, conduct fill up the wellbore to achieve slightly overbalance condition, 11. Shut down Foaming Pump 12. Pull Out of the hole the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly 13. If necessary, conduct waiting on Sand settle for 30 minutes to ensure well on clean condition and does not have any sand fill up again. 14. Rig Down Foam Unit	
14	**RUN UPPER SCREEN LINER ASSEMBLY -- FOLLOW SOP for SCREEN LINER** General Screen Liner Assembly With TBA STAND ALONE **Screen Liner slot size : 0.008"** **Note:** **For SASL Use Centralizer at upper most blank casing**	16

DON'T SLIP at Screen Body

Screen Liner Assembly depend on well program, design below is the basic concept.

ITEM (for Screen Liner Assembly -- should be STC Connection)	Length	Qty	TOP DEPTH (ft)
TBA 6-5/8" x 10-3/4"	1.05	1	267.0
6-5/8" Blank Liner x 30 ft	30	1	268.0
6-5/8" Screen Liner x 30 ft	30	3	298.0
6-5/8" Blank Liner x 20 ft	10	1	388.0
PMMS	1	1	398.0
Top Liner PK			399
Prod. Casing Shoe			500
Top of Screen Liner			298

15 **RUN LEAD SEAL ON UPPER SCREEN LINER, SET & BUMP DOWN**

Follow SOP for Run & Set Lead Seal

Check pin of Lead Seal -- remove 1 pin for shallow (>300 ft)

Use 6-8 jts DC to Bump Lead Seal.

Link :

https://collab001-idr.sp.chevron.net/sites/smodncwwc/ce/Wellview%20Related%20Files/SPECIFIC%20PROCEDURE%20-%20RUNNING%20AND%20SET%20LEAD%20SEAL.docx

2

16 **CLEAN OUT SAND TO PBTD BULL PLUG**

RIH with mule shoe and clean out sand to PBTD. Use Foam whenever required if there is experience loss circulation.

Ensure water supply is enough during clean out

Note,

Monitoring displacement process is needed to develop according to potential well control event while displaced water / brine with lower density fluid such foam. Below is procedure for foaming which contain of monitoring parameter and Kick mitigation.

1-3. Spot, Rig Up Foaming Unit. Fill up water tank.

4. Set Up line , conduct pressure test on discharge line of foaming unit on 1000 Psi

5. Add and mix the chemical (Foamer) into the water tank depend on wellbore requirement

6. Run in hole and lower the mule shoe / open ended/ rotary shoe and tubular Bottom hole assembly until 2-10 ft above top of sand

7. Install the inside BOPE

8. Conduct the foam / aerated pumping

	a. Pump the water using triplex pump on 1-3 BPM while monitor pressure gauge and control panel b. Pump the air slowly on 1300 – 1800 RPM while monitor pressure gauge and control panel c. Closed monitor the return (adjust the chemical foam until stable on return) d. Lower the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly slowly while monitor the return • If the tubular string does not utilize the / check valve/ float sub, conduct fill up the wellbore to achieve slightly overbalance condition before disconnect the current tubular for make up the next additional tubular string. • In the event that the return is stop (No Return on surface) , inform the driller to rise up the tubular into the safe area above the open interval / perforation • If foaming job is suspended for certain time, ensure to immediately secure the tubular string into the safe area above the open interval/perforation e. Continue the foam pumping and clean out until PBTD (clean well) , take sand sample from the wellbore on line manifold rig 9. Avoid environmental pollution by Spray the foam on breaker tank using de-foamer to reduce the enormous volume of foam on the breaker tank 10. After well confirm on clean condition, conduct fill up the wellbore to achieve slightly overbalance condition, 11. Shut down Foaming Pump 12. Pull Out of the hole the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly 13. If necessary, conduct waiting on Sand settle for 30 minutes to ensure well on clean condition and does not have any sand fill up again. 14. Rig Down Foam Unit Monitoring displacement process is needed to develop according to potential well control event while displaced water / brine with lower density fluid such foam. Below is procedure for foaming which contain of monitoring parameter and Kick mitigation. 1. Spot Foaming Unit 2. Rig Up Foaming Unit 3. Fill up water tank 4. Set Up line , conduct pressure test on discharge line of foaming unit on 1000 Psi 5. Add and mix the chemical (Foamer) into the water tank depend on wellbore requirement 6. Run in hole and lower the mule shoe / open ended/ rotary shoe and tubular Bottom hole assembly until 2-10 ft above top of sand 7. Install the inside BOPE 8. Conduct the foam / aerated pumping a. Pump the water using triplex pump on 1-3 BPM while monitor pressure gauge and control panel b. Pump the air slowly on 1300 – 1800 RPM while monitor pressure gauge and control panel c. Closed monitor the return (adjust the chemical foam until stable on return)	

	d. Lower the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly slowly while monitor the return • If the tubular string does not utilize the / check valve/ float sub, conduct fill up the wellbore to achieve slightly overbalance condition before disconnect the current tubular for make up the next additional tubular string. • In the event that the return is stop (No Return on surface) , inform the driller to rise up the tubular into the safe area above the open interval / perforation • If foaming job is suspended for certain time, ensure to immediately secure the tubular string into the safe area above the open interval/perforation e. Continue the foam pumping and clean out until PBTD (clean well) , take sand sample from the wellbore on line manifold rig 9. Avoid environmental pollution by Spray the foam on breaker tank using de-foamer to reduce the enormous volume of foam on the breaker tank 10. After well confirm on clean condition, conduct fill up the wellbore to achieve slightly overbalance condition, 11. Shut down Foaming Pump 12. Pull Out of the hole the mule shoe/ open ended/ rotary shoe and tubular Bottom hole assembly 13. If necessary, conduct waiting on Sand settle for 30 minutes to ensure well on clean condition and does not have any sand fill up again. 14. Rig Down Foam Unit	
17	**RIH WITH TUBING PUMP AS PER PROGRAM REQUIREMENT.** **PRESSURE TEST TUBING 300 PSI DURING 10 MINUTES. INSTALL TUBING DOUGHNUT.** RIH 2-3/4 TP - Regular and set @720ft. Note: Pressure test tubing hanger connection to MASP + 500 psi for 5 minutes	2
18	**INSTALL FOSV, N/D BOP, NU X-MAS TREE**	
19	**RIH PLUNGER, SUCKER ROD AND POLISH ROD. SET PLUNGER AS PER PUMP SET.** Perform pressure test Xmas tree (all tree, connections, stuffing box, or sample cock whichever applicable) to 500 psi + MASP as per stated in WSEA 10-L.	1

	Conduct test pumping 300 psi - 10 minutes. If it is OK- Continue R/D Rig, If leaking - Re POH Tubing Pump & check, change TP if needed	
20	**RDMO** Ensure to do Proper Handover with PMT. Note: Ensure to write down handover information in the tour report.	1
CONTINGENCY PLAN IF FOUND SIGN OF CASING LEAK/CLAY PROBLEM	**RIH WITH RETRIEVABLE PACKER AND CONDUCT PRESSURE TEST CASING -- follow SOP & procedure for Running, Setting & Testing.** Pressure Test casing with 300 PSI and hold for 5 mnts. IF there is any indication of casing leak, perform the following step: 1. Set the packer in the middle of the depth and perform pressure test casing to 300 psi and hold for 5 mnts to test the packer. 2. Locate casing leak with Pressure Washer (range accuration 10 ft) 3. Perform injectivity test with low rate and record as a base line for Squeeze Cementing Design 4. Prepare for next step (Squeeze Job)	4
	ISOLATE OPEN ZONE WITH ONE OF BELOW METHOD DEPEND ON WELL CONDITION: 1. Using packer-- Retrievable Packer + Dump 5 ft sand on top a. RIH with Retrievable Packer follow SOP for Running and Setting Packer complete with Setting/Releasing Tools. b. RIH with Mule Shoe and dump sand above packer. 2. Dump sand above TBA and Cement cap 5 ft a. RIH with Mole Shoe and dump sand above TBA/open interval or depend on program. b. Dump 5 - 10 ft Cement Cap with Class "G" Cement on top of the sand Discuss with TL Operation if other isolation alternative need to be proposed.	8
	PREPARE CEMENTING UNIT AND PERFORM SQUEEZE CASING LEAK JOB Discuss with TL Operation & Completion Engineer to determine squeeze method (basic method is bullhead)	8
	WOC WOC as per design and monitor cement surface sample	6
	DOC TO ORIGINAL TOP OF CEMENT. RETEST SQUEEZE CASING LEAK RESULT. **P**ressure test casing and increase pressure gradually (per 100 psi) to get 300 psi - 5 mnts hold. If still leaking, discuss with TL Operation and Completion Engineer if required additional squeeze or other alternative will be taken.	12

	TOTAL TIME : P10	**228**
	TOTAL TIME : P50	**234**
	TOTAL TIME : P90	**279**
NOTE: **If got problem operation, please discuss with TL Operation & WSM Specialist** **If any deviation from program should be discuss with Completion Engineer.**		